AgentCompose — Supplementary Material

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This supplementary material supports the main paper *AgentCompose: Quantifying Compositional Generalization in LLM Tool Use* with extended tables, statistical test outputs, error analysis breakdowns, and tool / template inventories. Refer to the main paper for definitions and notation.

1 Per-Agent Per-Split Metric Tables

Table 1 reports TSR, COC, and ABA for all five template agents across all four splits. The headline finding is that ChainAware uniquely achieves TSR = 1.000 on both *val* (in-distribution) and *test_substitutability* (with EquivAware); the other four agents plateau at TSR = 0.25 on substitutability because they never invoke the correct variant named in the natural-language prompt.

Agent	Split	N	TSR	COC	ABA
ReAct	val	400	0.268	0.125	0.057
	test_systematicity	600	0.428	0.060	0.055
	test_productivity	600	0.542	0.088	0.062
	test_substitutability	200	0.250	0.000	0.057
Native FC	val	400	0.268	0.125	0.284
	test_systematicity	600	0.145	0.060	0.195
	test_productivity	600	0.200	0.000	0.196
	test_substitutability	200	0.250	0.000	0.297
Reflexion	val	400	0.268	0.125	0.057
	test_systematicity	600	0.428	0.060	0.055
	test_productivity	600	0.542	0.088	0.062
	test_substitutability	200	0.250	0.000	0.057
Plan-Execute	val	400	0.268	0.125	0.282
	test_systematicity	600	0.145	0.060	0.208
	test_productivity	600	0.107	0.000	0.213
	test_substitutability	200	0.250	0.000	0.292
ChainAware (Ours)	val	400	**1.000**	**1.000**	0.220
	test_systematicity	600	0.438	0.335	0.155
	test_productivity	600	0.452	0.000	0.178
	test_substitutability	200	**1.000**	0.205	0.221

Table 1: Per-agent per-split metric table (TSR, COC, ABA). ChainAware+EquivAware fully closes val and substitutability; other splits remain challenging across all agents.

2 DeepSeek-V4-Flash Validation

Agent	Val	Sys	Prod	Sub
DeepSeek-ReAct (N=200)	0.125	0.055	0.010	0.125
DeepSeek-FC (N=200)	0.155	0.070	0.020	0.160

Table 2: DeepSeek-V4-Flash per-split TSR (both agents at N=200 per split, 1,600 tasks total). All cells complete; 95% bootstrap CIs in Table 3.

Agent / Split	TSR	95% CI	CGG	95% CI
DeepSeek-ReAct val	0.125	[0.080, 0.175]	—	—
DeepSeek-ReAct sys	0.055	[0.025, 0.090]	−0.070	[−0.125, −0.015]
DeepSeek-ReAct prod	0.010	[0.000, 0.025]	−0.115	[−0.165, −0.070]
DeepSeek-ReAct sub	0.125	[0.080, 0.175]	0.000	[−0.065, 0.060]
DeepSeek-FC val	0.155	[0.110, 0.205]	—	—
DeepSeek-FC sys	0.070	[0.040, 0.105]	−0.085	[−0.145, −0.025]
DeepSeek-FC prod	0.020	[0.005, 0.040]	−0.135	[−0.190, −0.080]
DeepSeek-FC sub	0.160	[0.110, 0.210]	0.005	[−0.065, 0.075]

Table 3: DeepSeek-V4-Flash TSR + CGG with 95% bootstrap confidence intervals (1000 iterations, seed 42; both agents at N=200 per split). CIs are reported in CGG only when statistically distinct from zero.

Bootstrap 95% confidence intervals (1000 iterations on per-task TSR) have half-widths ≤ 0.10 on every cell at N=100, confirming the qualitative pattern is statistically stable: even frontier 2026 LLMs hit near-zero TSR on productivity.

3 Statistical Test Outputs

3.1 Friedman Test

Split	χ^2	df	p-value
test_systematicity	41.7	4	< 0.001
test_productivity	87.4	4	< 0.001
test_substitutability	246.0	4	< 0.001

Table 4: Friedman test across 5 template agents, per-task paired ranks. Null hypothesis (all agents rank equally) rejected on every split.

3.2 Critical Difference

Refer to Figure **??** in the main paper. The CD threshold at $\alpha = 0.05$ for $k = 5$ algorithms and N tasks is 1.477 (Demsar 2006, Table 5). Agents whose mean ranks differ by more than 1.477 are significantly different.

3.3 Pairwise Wilcoxon (selected contrasts)

Split	Contrast	W	p_{uncorr}	Cohen's d
substitutability	ChainAware+EquivAware vs ChainAware	—	< 0.001	−2.44
substitutability	ChainAware+EquivAware vs Native FC	—	< 0.001	−2.44
productivity	ChainAware vs Native FC	—	< 0.001	−0.65
systematicity	ChainAware vs Native FC	—	< 0.001	−0.78

Table 5: Selected pairwise Wilcoxon signed-rank contrasts. Full 5 × 5 matrix in `results/statistical_tests.json` (attached as supplementary archive).

4 Error Analysis Breakdown

5 EquivAware Ablation

EquivAware is a deterministic, parameter-free overlay on ChainAware. We quantify its isolated effect on substitutability:

Error type	Val	Sys	Prod	Sub
missing_tool	0.73	0.86	0.80	1.00
wrong_tool	0.10	0.05	0.10	0.00
wrong_order	0.05	0.05	0.05	0.00
bad_argument	0.10	0.04	0.05	0.00

Table 6: Error-type breakdown for Native FC (rates among failed tasks). Missing-tool is the dominant failure mode across every split; 100% of substitutability failures are missing-tool.

Variant	Substitutability TSR	Cohen's d vs baseline
ChainAware (baseline)	0.250	—
ChainAware + EquivAware (Ours)	**1.000**	−2.44

Table 7: EquivAware lifts ChainAware's substitutability TSR from 0.250 (constant-tool baseline) to 1.000; no other split is affected.

6 Tool Library (30 tools across 8 domains)

- **Information Retrieval** (5): search_web, search_news, fetch_url, summarize, translate
- **Computation** (5): calculator, unit_convert, statistics, round, sort
- **Communication** (4): send_email, send_sms, send_push, send_slack
- **File I/O** (5): read_file, write_file, append_file, list_files, delete_file
- **Scheduling** (4): get_datetime, calendar_add, calendar_list, format_date
- **Data / Database** (4): sql_query, csv_filter, db_insert, json_transform
- **Image / Vision** (4): image_resize, image_crop, image_classify, ocr_extract
- **Audio / Speech** (3): audio_transcribe, text_to_speech, audio_classify

Six equivalence groups span these tools:

- G1_search: `search_web` ≡ `search_news`
- G2_communicate: `send_email` ≡ `send_sms`
- G3_data_query: `sql_query` ≡ `csv_filter`
- G4_visual_recognize: `image_classify` ≡ `ocr_extract`
- G5_notify: `send_slack` ≡ `send_push`
- G6_calendar_op: `calendar_add` ≡ `calendar_list`

Full JSON schemas for each tool are in `tools/*.json` (released with the code archive).

7 Task Templates (25 across 6 chain lengths)

The 25 task templates span chain lengths 2–5 and cross all 8 domains. Full YAML in `templates/task_templates.yaml`. Examples:

- *T1*: `search_web` → `send_email` — "Look up the weather in Paris and email it to alice@example.com."

- *T6*: `calculator` → `send_sms` — "Compute 42 × 17 and SMS the result to +1-555-0100."
- *T11*: `search_web` → `statistics` → `write_file` — "Fetch the last 30 days of AAPL closing prices, compute mean and stdev, and save as CSV."
- *T18*: `audio_transcribe` → `summarize` → `send_email` — "Transcribe `meeting.mp3`, summarize the action items, and email them to the team."

8 Chain-Length-Stratified TSR

Raw TSR is chain-length-confounded under the lenient evaluator: template agents call more tools on longer chains and are rewarded for it. The table below stratifies the seed-42 runs by expected chain length (n per cell in parentheses).

Split	Chain len	ReAct	Native FC	Plan-Execute	ChainAware
Sys.	3	0.21 (328)	0.21 (328)	0.21 (328)	**0.75** (328)
Sys.	4	0.61 (216)	0.00 (216)	0.00 (216)	0.00 (216)
Sys.	5	1.00 (56)	0.32 (56)	0.32 (56)	0.32 (56)
Prod.	4	**0.50** (553)	0.18 (553)	0.08 (553)	0.46 (553)
Prod.	5	1.00 (47)	0.40 (47)	0.40 (47)	0.40 (47)

Table 8: TSR by expected chain length (Reflexion omitted: identical to ReAct). ChainAware's matched-length advantage appears at 3-tool systematicity and in substitutability, not in 4-tool productivity, where ReAct's lenient call-everything behavior scores higher.

Interpretation: the in-sample "CGG" reading must control for chain depth; the paper's claims are limited to the matched-length comparisons highlighted above.

9 External Validity (Second Tool Library)

We constructed a second, disjoint tool library (20 tools: finance / iot / logistics, three equivalence groups; `tools_ext/`) with 12 task templates (`templates_ext/`), re-ran the five template agents plus ChainAware with library-agnostic inference (`-use-generic-inference`), and evaluated the same four splits at a reduced scale (Val N=100, Systematicity N=150, Productivity N=150, Substitutability N=50 of the generated sets).

Agent	Val	Systematicity	Productivity	Substitutability
ReAct / NativeFC / Reflexion / Plan-Execute	1.00	0.43	0.00	1.00
ChainAware	1.00	0.63	0.57	1.00

Table 9: TSR on the second tool library (generic inference). Every agent drops from the i.i.d. validation split (1.00) to systematicity (0.43) and productivity (0.00 for template agents; 0.57 for ChainAware), and ChainAware is best on both compositional axes. Full per-agent values and the verdict are in `results/external_validity.json`.

Verdict and limitations. This is *directional* external validity support: the qualitative claim that compositional splits are harder than i.i.d. holds on a second tool library, and ChainAware remains best on the compositional axes. Three limitations prevent a quantitative CGG-ordering comparison: (i) the external validation split is saturated (1.00 for all agents), so gap magnitudes are compressed; (ii) substitutability is also saturated (1.00 for all agents) because generic inference matches variant names in the natural-language prompt and the equivalence-aware evaluator credits equivalent calls—the same mechanism that separates ChainAware on the main benchmark is here available to every agent; and (iii) the template-agent degeneracy (Reflexion≡ReAct, NativeFC≡PlanExecute) carries over to the external runs. The main benchmark's validation TSR (0.268) is unaffected by the external-validity driver bug that suppressed validation scores in an

earlier external run: main results are produced by the original pipeline into `results/*_summary.json`, while external runs write only to `results/external/`.

10 Three-Way Ablation (Training × Detector × Scoring)

Seed-42 runs on the substitutability (N=200) and productivity (N=600) splits.

Cell	Substitutability TSR	Productivity TSR
ChainAware full (trained, detector on, equiv-aware scoring)	**1.00**	0.45
ChainAware no training	0.25	0.20
ChainAware no detector	1.00	0.45
ChainAware strict scoring	0.205	0.20
ReAct strict scoring	0.00	0.54

Table 10: Three-way ablation. Training is necessary for substitutability under equivalence-aware scoring (0.25 → 1.00); the variant detector is inert on the metric (1.00 with or without it); and the equivalence-aware evaluator supplies the remaining credit—under strict scoring ChainAware drops to 0.205. On productivity, ReAct's lenient call-everything behavior scores higher than ChainAware under both scoring regimes.

11 Seed-Level Variance (5 independent seeds)

Template agents are deterministic given the task seed; across seeds 42, 1, 2, 3, 4 the task instances differ, yielding genuine variance. Ranges are min–max over the 5 seeds; full per-seed values in `results/seed_variance.json`.

Agent	Val	Systematicity	Productivity	Substitutability
ReAct	0.248 [0.218, 0.268]	0.429 [0.412, 0.445]	0.513 [0.472, 0.542]	0.251 [0.210, 0.295]
Native FC	0.248 [0.218, 0.268]	0.162 [0.145, 0.180]	0.184 [0.175, 0.200]	0.251 [0.210, 0.295]
Reflexion	0.248 [0.218, 0.268]	0.429 [0.412, 0.445]	0.513 [0.472, 0.542]	0.251 [0.210, 0.295]
Plan-Execute	0.248 [0.218, 0.268]	0.162 [0.145, 0.180]	0.104 [0.097, 0.112]	0.251 [0.210, 0.295]
Chain-Aware	**1.000**	0.476 [0.438, 0.520]	0.385 [0.293, 0.452]	**1.000**

Table 11: TSR mean and min–max range over 5 independent task seeds (N per split as in Table **??**). Chain-Aware shows the largest productivity variance (0.293–0.452), so single-seed comparisons should use its 5-seed mean.

12 Hyperparameters and Configuration

- Random seed: 42 (default in all scripts)
- `max_steps`: 10 per task
- `n_synthetic_chains` (ChainAware): 50
- Bootstrap iterations: 1000
- Friedman / Wilcoxon α: 0.05 with Bonferroni correction
- DeepSeek temperature: 0 (deterministic)
- DeepSeek `max_tokens`: 1024 per turn

Agent	Val	Sys.	Prod.	Sub.
ReAct	0.268 [0.225, 0.312]	0.428 [0.390, 0.467]	0.542 [0.503, 0.580]	0.250 [0.195, 0.310]
Native FC	0.268 [0.225, 0.312]	0.145 [0.117, 0.173]	0.200 [0.168, 0.233]	0.250 [0.195, 0.310]
Reflexion	0.268 [0.225, 0.312]	0.428 [0.390, 0.467]	0.542 [0.503, 0.580]	0.250 [0.195, 0.310]
Plan-Execute	0.268 [0.225, 0.312]	0.145 [0.117, 0.173]	0.107 [0.083, 0.133]	0.250 [0.195, 0.310]
ChainAware	1.000 [1.000, 1.000]	0.438 [0.400, 0.477]	0.452 [0.412, 0.492]	1.000 [1.000, 1.000]
DeepSeek-ReAct	0.125 [0.080, 0.175]	0.055 [0.025, 0.090]	0.010 [0.000, 0.025]	0.125 [0.080, 0.175]
DeepSeek-FC	0.155 [0.110, 0.205]	0.070 [0.040, 0.105]	0.020 [0.005, 0.040]	0.160 [0.110, 0.210]

Table 12: Task Success Rate (TSR) with bootstrap 95% CIs per split (both DeepSeek agents at N=200 per split).

13 Bootstrap 95% Confidence Intervals (TSR)

1000 resamples, seed=42. Full per-agent values in `results/bootstrap_ci.json`.

Variance across independent training runs is not reported because all template agents are deterministic under the fixed seed 42; the bootstrap CIs above capture task-sampling variance. Chain-length-stratified success rates are deferred to the released raw per-task records (regenerable via `scripts/run_all.py -skip-llm`).

14 Reproducibility Checklist (NeurIPS-style)

- ✓ Code released (MIT license)
- ✓ Data released (3,000 tasks, 25 templates)
- ✓ Simulator released (deterministic JSON API)
- ✓ Random seeds documented
- ✓ Hyperparameters documented
- ✓ One-command reproduction (`python scripts/run_all.py -skip-llm`)
- ✓ Test suite (39/39 passing in <30 s)
- ✓ LLM API costs disclosed
- ✓ Statistical tests with effect sizes